Suman Maiti

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Indian Institute of Technology Kharagpur

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Google Scholar | GitHub | LinkedIn

Research Interests

Security of cyber-physical systems with a focus on smart grids: attack synthesis and vulnerability analysis, formal verification of learning-enabled controllers (Metric Temporal Logic, SMT solving, falsification), and safe reinforcement learning for attack mitigation.

Education

- **Indian Institute of Technology Kharagpur** — Kharagpur, India
  Doctor of Philosophy (PhD) - Computer Science and Engineering — *January 2022 – Present*
  Advisor: Prof. Soumyajit Dey **Area:** Security of cyber-physical systems
  Courses: High Performance Computer Architecture, Foundations of Algorithm Design and Machine Learning, Artificial Intelligence for Cyber-Physical Systems, Computational Foundations of Cyber-Physical Systems, Statistical Methods, English for Technical Writing.
- **Kalyani Government Engineering College** — Nadia, India
  Bachelor of Technology - Electrical Engineering; CGPA: 9.03/10 — *September 2017 – August 2021*
  Courses: Digital Electronics, Analog Electronics, Control Systems, Signals and Systems, Power Systems, Power Electronics.
- **D.A.V. Model School IIT Campus Kharagpur** — Kharagpur, India
  All India Senior School Certificate Examination, Class 12; Percentage: 93.8% — *March 2017*
- **D.A.V. Model School IIT Campus Kharagpur** — Kharagpur, India
  All India Secondary School Examination, Class 10; CGPA: 10/10 — *March 2015*

Publications

Peer-Reviewed

[1] **Mitigating Cyber-Physical Attacks in Smart Grids: A Learner Verifier Architecture**
Suman Maiti, Sunandan Adhikary, Somnath Hazra, Soumyajit Dey
52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), 2026. (Accepted)

[2] **Game-Theoretic Adaptive Voltage Threshold Tuning Against HVAC Based Load Demand Variations**
Suman Maiti, Anjana B., Sunandan Adhikary, Soumyajit Dey
52nd Annual Conference of the IEEE Industrial Electronics Society (IECON), 2026. (Accepted)

[3] **A Learning-Enabled Framework for Profit-Driven Attack Synthesis in Electricity Markets**
Suman Maiti, Anirban Das, Soumyadyuti Ghosh, Soumyajit Dey, Debdeep Mukhopadhyay
International Conference on Security, Privacy and Applied Cryptographic Engineering (SPACE), 2026. (Accepted)

[4] **Differentially Private Real Time Pricing Control for Smart Grids**
Soumyadyuti Ghosh, **Suman Maiti**, Soumyajit Dey, Debdeep Mukhopadhyay
ACM Transactions on Cyber-Physical Systems, 2025.

[5] **Targeted Attack Synthesis for Smart Grid Vulnerability Analysis**
Suman Maiti, Anjana B., Sunandan Adhikary, Ipsita Koley, Soumyajit Dey
ACM SIGSAC Conference on Computer and Communications Security (CCS), 2023.

[6] **Symmetrical and Asymmetrical Fault Studies of Wind-Farm Embedded Power Network**
Suman Maiti, Suvodip Som, Soumen Mondal, Pritam Kumar Gayen
IEEE Devices for Integrated Circuit (DevIC), pp. 330–334, 2021.

Under Review

[R1] **A Multivariate Attack Vector Synthesis Framework for Enhancing Smart Grid Vulnerability Analysis**
Suman Maiti, Anjana B., Sunandan Adhikary, Ipsita Koley, Ashish Ranjan Hota, Soumyajit Dey
ACM Transactions on Cyber-Physical Systems (major revision). arXiv:2306.09057.

In Preparation

[W1] **A Variance-Optimal Replay Framework for Safe Off-Policy Reinforcement Learning**
Suman Maiti, Sunandan Adhikary, Somnath Hazra, Soumyajit Dey
Manuscript in preparation.

Research Experience

- **PhD Research Scholar | Advisor: Prof. Soumyajit Dey**
  Department of Computer Science and Engineering, IIT Kharagpur *January 2022 – Present*
  - **Smart Grid Security**: Attack synthesis, formal verification and safe reinforcement learning for cyber-physical attack analysis and mitigation in power grids. Developed a smart grid hardware-in-the-loop setup (OP4510, SEL-421) at SHADES Laboratory.
- **Junior Research Fellow | Guidance: Prof. Soumyajit Dey**
  Department of Computer Science and Engineering, IIT Kharagpur *October 2021 – December 2023*
  - **Smart Grid Privacy and Security (SERB-funded)**: Studied vulnerabilities in smart grid environments and their countermeasures, applying reinforcement learning with the falsification tool S-TaLiRo to IEEE 9, 14, 37 and 39 bus models.
- **Summer Intern | Guidance: Prof. Pabitra Mitra**
  Department of Computer Science and Engineering, IIT Kharagpur *June 2019 – July 2019*
  - **Design of digital clock using 8085 microprocessor**: Programmed the 8085 microprocessor to display time on a seven-segment display.

Teaching Assistantship

Department of Computer Science and Engineering, IIT Kharagpur

- **Programming and Data Structures Laboratory** – Prof. Dipanwita Roy Chowdhury — Spring 2026
- **Reinforcement Learning** – Prof. Soumyajit Dey — Autumn 2025
- **Formal Language and Automata Theory** – Prof. Soumyajit Dey, Prof. Animesh Mukherjee — Spring 2025
- **Programming and Data Structures (Theory)** – Prof. Soumyajit Dey — Autumn 2024
- **Formal Language and Automata Theory** – Prof. Soumyajit Dey, Prof. Animesh Mukherjee — Autumn 2023 – Spring 2024
- **Programming and Data Structures Laboratory** – Prof. Pallab Dasgupta — Spring 2023
- **Programming and Data Structures Laboratory** – Prof. Soumyakanti Ghosh, Prof. K. S. Rao — Autumn 2022
- **Programming and Data Structures Laboratory** – Prof. Debasis Samanta — Spring 2022

Mentoring

- **AICTE Faculty Quality Improvement Programme**: Guided five faculty members on applications of reinforcement learning in cyber-physical systems operation: Dr. Barnali Kundu (GNIT), Dr. Aditi Chatterjee (C.V. Raman Global University), Dr. R. Sireesha and Dr. Kumar K (Sri Venkateswara College of Engineering, Tirupati), Mr. Pritam Chandra (Meghnad Saha Institute of Technology).
- **Students**: Harsh Khashbage (B.Tech thesis, IIT Kharagpur); Soumojit Chatterjee and Kaushal Kumar Singh (dual-degree projects, IIT Kharagpur); Anirban Das (research intern, Jadavpur University, May–July 2025).

Professional Service

- **Journal Reviewer**: IEEE Transactions on Power Systems; IEEE Transactions on Smart Grid; IEEE Transactions on Industrial Cyber-Physical Systems; IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems.
- **Conference Reviewer**: ICCPS 2024, 2025, 2026; AAAI 2023; Euromicro DSD 2023, 2024; IEEE RTSS 2023.
- **Artifact Evaluation Committee**: ICCPS 2026.

Awards, Grants and Talks

- **CHANAKYA PhD Fellowship**, AI4ICPS Hub, IIT Kharagpur — 2024 – 2025
- **ACM CCS Student Travel Grant** — 2023
- **ACM IARCS Travel Grant** (INR 1,00,000) — 2023
- **Invited Talk**, ACM ARCS Event, NISER Bhubaneswar: smart grid security and attack synthesis — 2023

- **GATE** (Electrical Engineering), 94.4 percentile 2021

Projects

- **Formal Methods for Smart Grid Privacy and Security**: Vulnerability analysis of IEEE 9, 14 and 39 bus generation and distribution systems using a reinforcement learning agent and the formal falsification tool S-TaLiRo.
- **Fault Studies of Wind Farm Embedded Power Network**: Simulation of symmetrical and asymmetrical faults in a 225/90 kV power network; analysis of transmission-line voltage and power waveforms.
- **CPU Architecture Performance Estimation using gem5**: Generation of CPU configurations from user-specified parameters and evaluation with benchmark programs on the gem5 simulator.
- **Autonomous Driving on an F1/10 Vehicle**: Controller design for track following with obstacle avoidance.

Technical Skills

- **Languages**: Python, C++, C, PHP, LaTeX
- **Machine Learning**: PyTorch, TensorFlow, Keras, Scikit-learn, Stable-Baselines3, Gymnasium
- **Formal Methods**: Z3 SMT Solver, S-TaLiRo, Metric Temporal Logic, Hybrid Automata
- **Power System Tools**: MATLAB, Simulink, RT-LAB, LabVIEW, ARTEMiS, HYPERSIM, ePHASORsim, eMEGAsim, EMTP-RV, pandapower
- **Hardware**: OP4510 (OPAL-RT), SEL-421 (Schweitzer Engineering Laboratories), Raspberry Pi
- **ICS Protocols**: DNP3, DNP3 Secure Authentication, Modbus RTU/TCP
- **Platforms and Tools**: Linux, Windows, Git, Docker, Jupyter, gem5